Stephen James Kranz

SCIENTIST AND DESIGNER · BERKELEY, CALIFORNIA

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portfolio.calipercrayon.com

Designer with an M.S. in materials science and engineering. I make books, scientific illustrations, branding and explainer videos for research and education on scientific and engineering topics.

EXPERIENCE

Caliper & Crayon · Founder and designer — 2020 – now

- Designed the Magical Flag of Peace and produced 671 hand-made variations of it, photographed and cataloged in a filterable web archive I built.
- Wrote a science highlight of an Advanced Light Source paper with five original diagrams, reviewed for accuracy by the lead author.
- For GaeaStar, built out a ceramics lab, wrote its safety protocols, and prototyped and documented a glaze application technique (2023).
- Built a Canva brand kit and slide template for the Caliper & Crayon studio.

Origin Laboratories · Lab manager, scientist — 2017 – 2020

- Managed an R&D 3D-printing lab of 40 printers and led a team of two to three scientists and technicians.
- Developed procedures for using the printers in industrial mass manufacture, evaluated third-party resins, and produced sample parts for customers.

Autodesk · Visual storyteller, researcher — 2016

- Wrote, directed, filmed, illustrated and edited customer stories and feature explainers for the Ember 3D printer, within Autodesk brand standards.
- Researched print speed and quality; co-inventor on a patent for printing sharper vertical edges.

EngineerGuy · Video producer with Prof. Bill Hammack — 2013 – 2015

- Co-wrote, directed, filmed, illustrated, animated and edited explainer videos with tens of millions of views, including one with over 19 million.
- Photographed and designed a companion book to a series of videos on Albert Michelson's Harmonic Analyzer (2014).

University of Illinois · Graduate researcher, Lewis lab — 2010 – 2013

- Created photographs and figures for seven papers, including the inside front cover of *Advanced Materials*; co-author on the cover paper.
- Directed, filmed, edited and animated a video on the lab's silver ink for *Chemical & Engineering News* (about 250,000 views).

TOOLS

InDesign, Illustrator, Photoshop, Cinema 4D, Canva, HTML and CSS, Webflow

SKILLS

Book and editorial design · typography and type design · scientific illustration · brand templates and identity · print production · photography · motion graphics · science writing

SELECTED WORK

Archie: Gold in the Fire
Typeset, designed and published, 2019

Albert Michelson's Harmonic Analyzer
Photographed and designed, 2014

EDUCATION

M.S., Materials Science & Engineering
University of Illinois, 2013

B.S., Materials Science & Engineering
Arizona State University, 2010

PATENTS

Enhanced three-dimensional printing of vertical edges
US 10,780,641 B2, 2020

Multinozzle deposition system for direct write applications
WO 2013/006399 A2, 2013